



ONYX-01 Concept Prototype

Simulating Design, Energy, and Logic through AI and Mathematical Modeling.

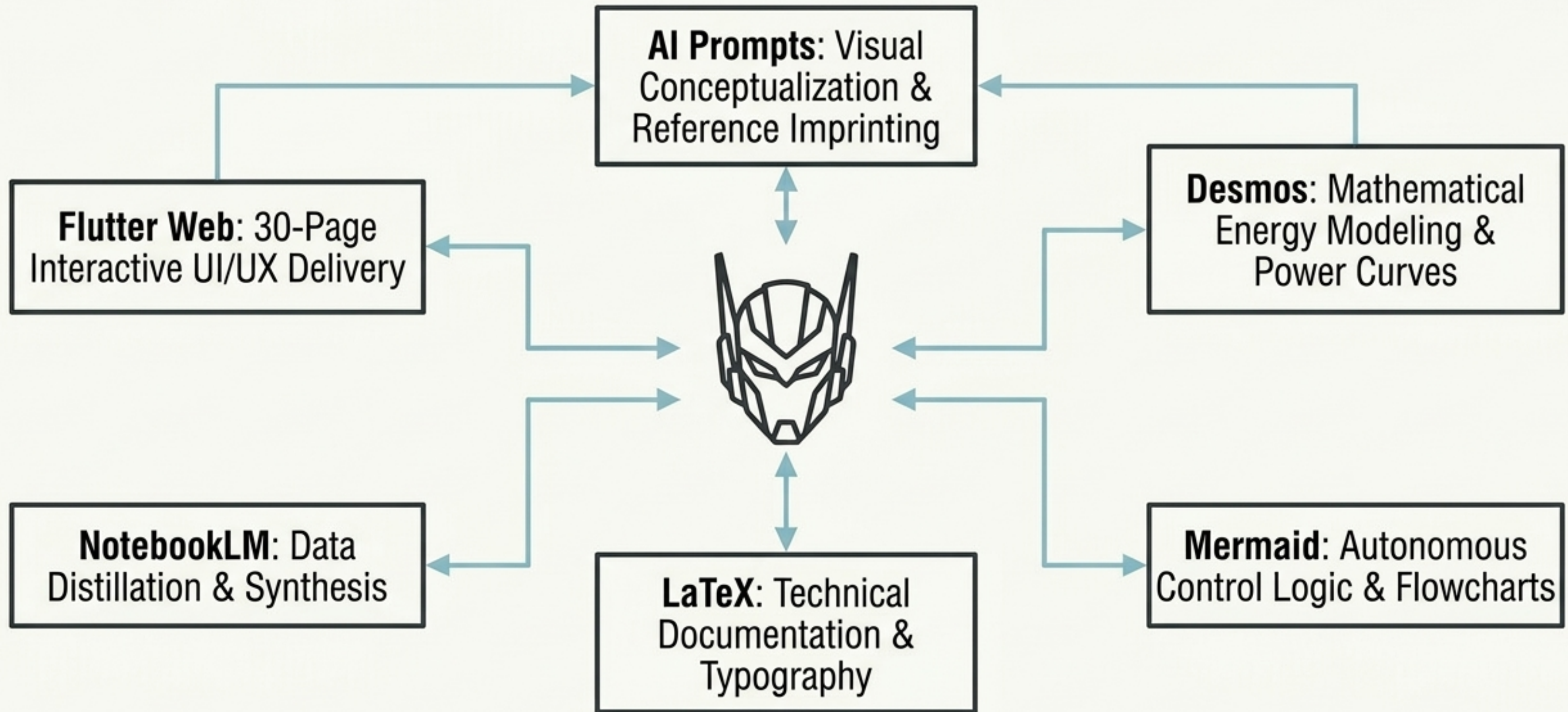


Defining the Physical Blueprint

- Matte black layered armor over graphite metallic skin
- Dual crest helmet with luminous pale blue optics
- V-shaped chest chassis
- Shoulder-mounted all-terrain wheels

CONCEPTUAL STATUS: Exploratory prototype only. No physical units exist. All subsequent data is digitally simulated.

The Digital Simulation Stack



Visual Design: The Limits of Zero-Shot Prompting

INPUT: "Black robot."



ZERO-SHOT RESULT: AI automatically assumed the scene, shape, lighting, and embedded generic text. The result lacks specific functional intent.

INPUT: Added specific model names, layered armor characteristics, eye color, and explicit layout constraints.



REFINED RESULT: High-fidelity, specific concept matching the ONYX-01 vision.

Cross-Domain Reference Imprinting

SOURCE: AUTOMOTIVE DESIGN



Extracted aggressive, angular neon lighting profiles.

SOURCE: INDUSTRIAL TIMEPIECE



Extracted matte black layered textures and metallic finishes.

TARGET: ONYX-01 ROBOTIC CONCEPT



TAKEAWAY:

Supplying cross-domain reference images establishes rigid boundaries, anchoring the fictional concept in real-world industrial design language.

Energy Modeling & Power Consumption

THE MATH

CAPACITY BASE:

$$E = 2400 \text{ Wh}$$

POWER CURVE:

$$P(v) = 200 + 0.5v^2 \text{ W}$$

ENDURANCE MODEL:

$$t(v) = E/P(v) \text{ hrs}$$

Note: Initial linear trend models ($t(v) = 12 - 0.15v$) were upgraded to the quadratic curve for simulated physics accuracy.

THE REALITY CHECK GRID

IDLE (0 km/h) -> **12.0 Hours**

CRUISING (20 km/h) -> **6.0 Hours**

ACTIVE (40 km/h) -> **2.4 Hours**

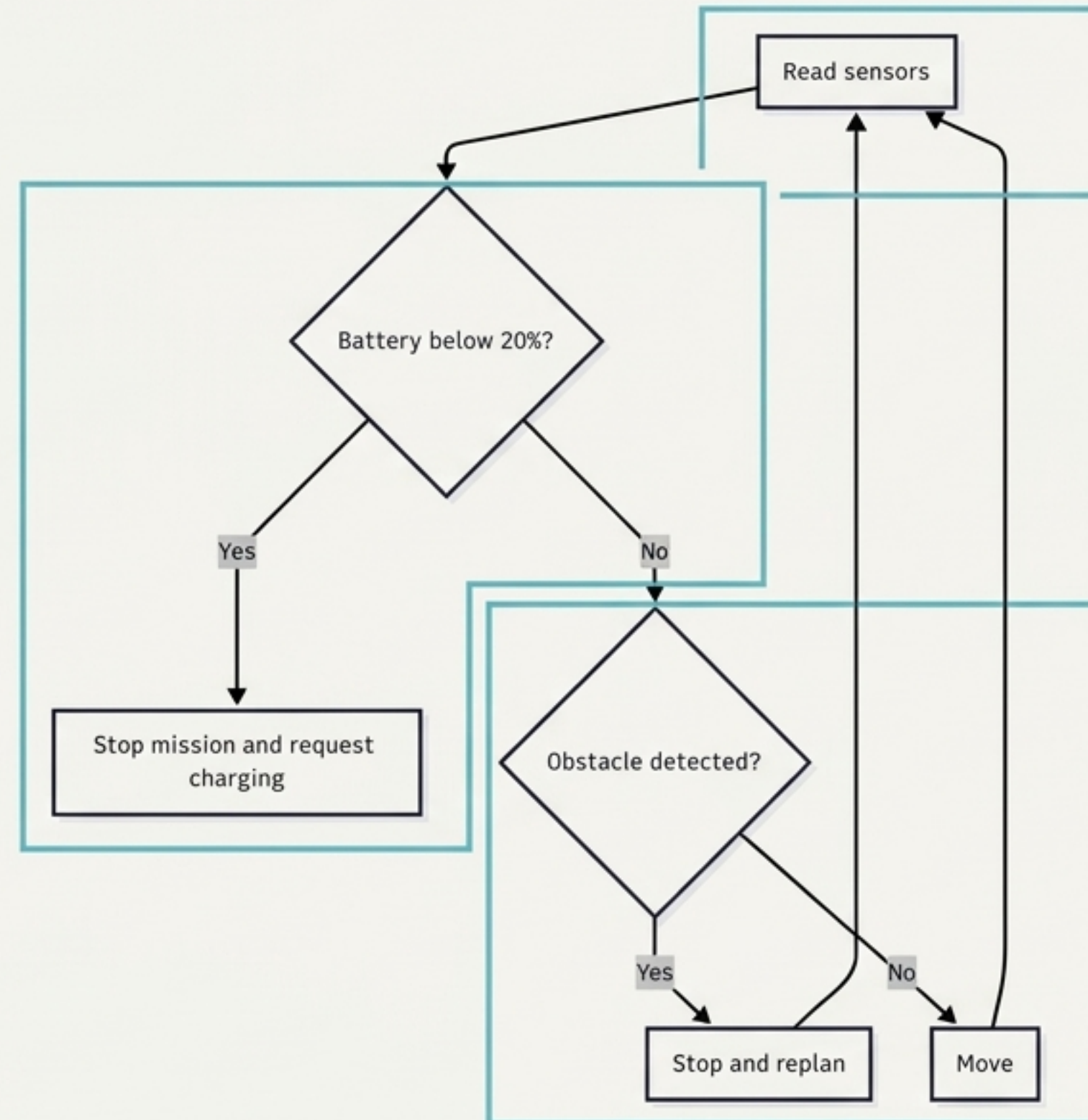
MAX SPEED (60 km/h) -> **1.2 Hours**

Autonomous Control Logic

PHASE 1: SENSE - Continuous ingestion of environmental and internal telemetry.

PHASE 3: NAVIGATE - Environmental interaction. Halt and replan, or execute movement and return to Phase 1.

PHASE 2: SURVIVE - Critical threshold check. If triggered, immediate mission abort.



Meta-Learning: The Science of the Prompt

THE MISCONCEPTION

Adding more descriptive text makes a prompt Few-Shot.

False. Highly detailed text is still fundamentally Zero-Shot if it lacks structural examples.

THE REALITY

True Few-Shot requires rigid boundaries.

True Few-Shot prompting requires providing explicit structural examples or format references to establish strict boundaries for the model's output.

Physical Constraints & Reality Checks

WARN_01 [MECHANICAL]: Transformation mechanisms are purely conceptual and lack proven engineering physics.

WARN_02 [ENVIRONMENTAL]: The current energy model exists in a vacuum. It entirely ignores incline gradients, payload weight, temperature variances, and friction loss.

WARN_03 [VISUAL]: AI-generated renders contain inherent structural continuity errors upon close inspection.



Bridging Imagination and Rigor

ONYX-01 serves as a foundational case study in modern workflows. By chaining visual AI, mathematical modeling, and logical structuring, we can stress-test theoretical concepts long before physical prototyping begins.

End of Simulation.